IGCSE 0607 Practice Questions — E5.5

Angles

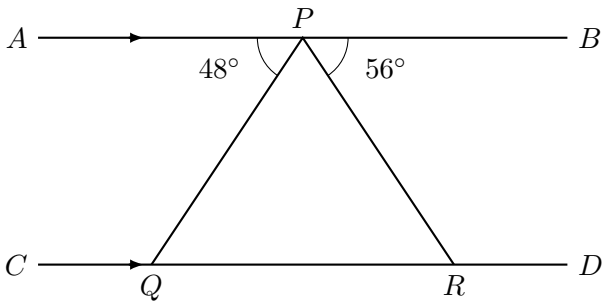
Total Questions: 21 — **Total Marks:** 135 Marks

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- **E5.5** Angles
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

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1 In the diagram, the line AB is parallel to the line CD.
P is a point on AB.
Q and R are points on CD.
The line PQ and the line PR are drawn.
Angle $APQ = 48°$ and angle $BPR = 56°$.



NOT TO SCALE

(a) Write down the value of angle PQR.

Angle $PQR =$ ..... ° **[1]**

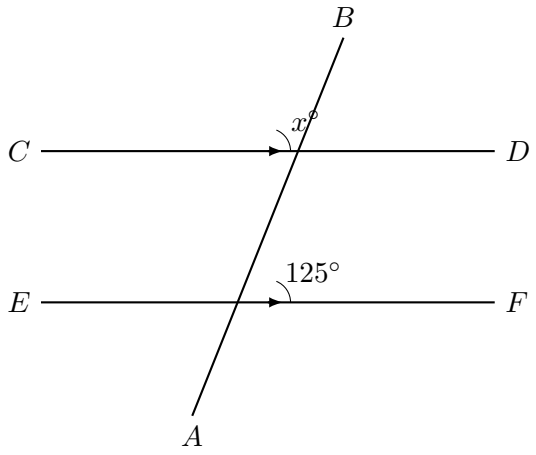
(b) Write down the value of angle PRQ.

Angle $PRQ =$ ..... ° **[1]**

(c) Work out the value of angle QPR.

Angle $QPR =$ ..... ° **[1]**

2 The diagram shows a straight line AB intersecting two parallel lines, CD and EF.



(a) Find the value of x.

$x =$ ..... **[1]**

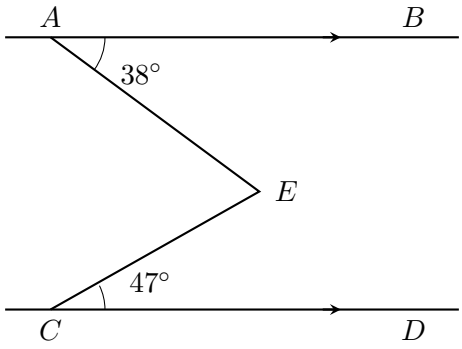
(b) Give a geometric reason for your answer to part (a).

..... **[1]**

(c) Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

3 In the diagram, the line AB is parallel to the line CD.
The point E lies between the two parallel lines.



NOT TO SCALE

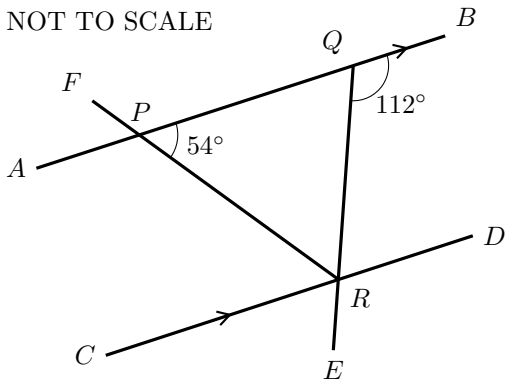
(a) Calculate the size of angle AEC.

Angle AEC = ... ° **[2]**

(b) A regular polygon has an exterior angle of 24°.
Work out the number of sides of this polygon.

..... **[2]**

4 In the diagram, AB is parallel to CD.
The straight line FPR intersects AB at P and CD at R.
The straight line QRE intersects AB at Q and CD at R.
Angle $QPR = 54°$ and angle $BQR = 112°$.



(a) Write down the size of angle PRC and give a geometrical reason for your answer.

Angle $PRC =$ ..... , Reason: ..... **[2]**

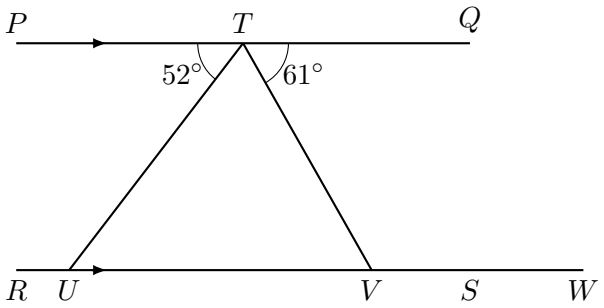
(b) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[2]**

(c) Write down the size of angle CRE.

Angle $CRE =$ ..... ° **[1]**

5 In the diagram, the line PQ is parallel to the line RS.
The point T lies on PQ, and the points U and V lie on RS.
The line RS is extended to a point W.
Angle $PTU = 52°$ and angle $QTV = 61°$.



(a) Calculate angle TUV.

Angle TUV = ... ° **[1]**

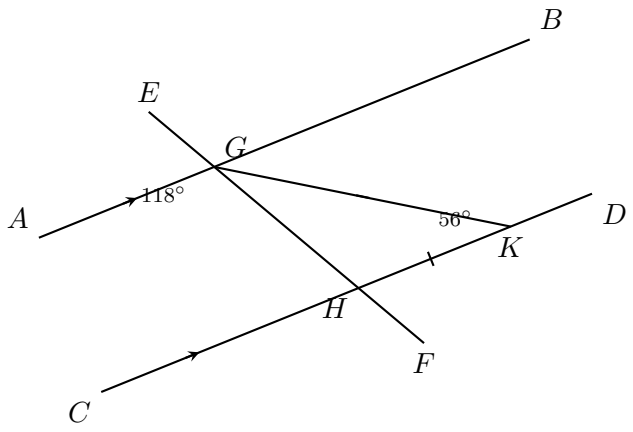
(b) Calculate angle UTV.

Angle UTV = ... ° **[2]**

(c) Calculate angle TVW.

Angle TVW = ... ° **[2]**

6 In the diagram, AB is parallel to CD.
The straight line EF intersects AB at G and CD at H.
Point K lies on CD such that $GK = HK$, $\angle AGH = 118°$ and $\angle GKH = 56°$.



(a) Find the size of angle GHD, giving a geometrical reason for your answer.

$\angle GHD =$ ..... ° **[2]**

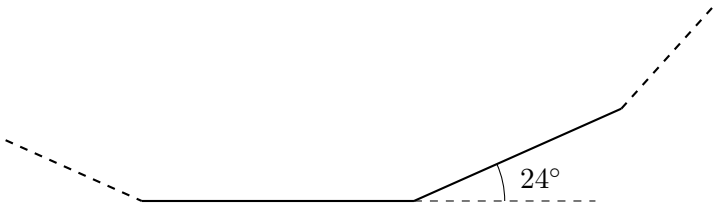
(b) Calculate the size of angle GHK.

$\angle GHK =$ ..... ° **[1]**

(c) Calculate the size of angle BGK.

$\angle BGK =$ ..... ° **[2]**

7 The diagram shows part of a regular polygon with n sides.
The exterior angle of this polygon is 24°.



(a) Calculate the value of n.

$n =$ ..... **[2]**

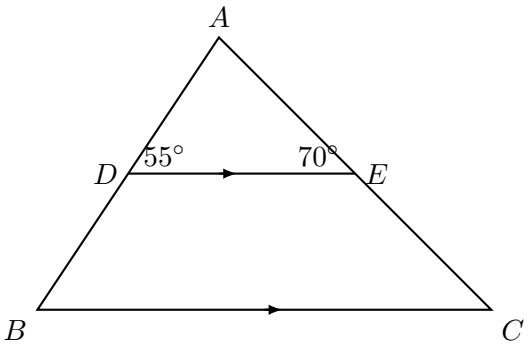
(b) Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

(c) A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

8 The diagram shows a triangle ABC with a line DE drawn parallel to BC.
D lies on AB and E lies on AC.
Angle $ADE = 55°$ and angle $AED = 70°$.



(a) Calculate angle DAE.

Angle $DAE =$ ..... ° **[2]**

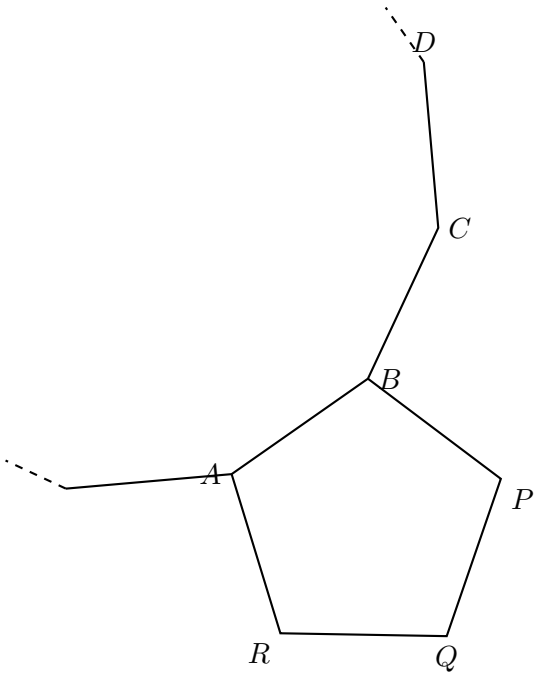
(b) Calculate angle ABC.

Angle $ABC =$ ..... ° **[1]**

(c) Explain why triangle ADE is an isosceles triangle.

[2]

9 The diagram shows part of a regular n-sided polygon $ABCDE\ldots$ and a regular pentagon $ABPQR$ sharing the common side AB.



(a) Calculate the size of one interior angle of the regular pentagon $ABPQR$.

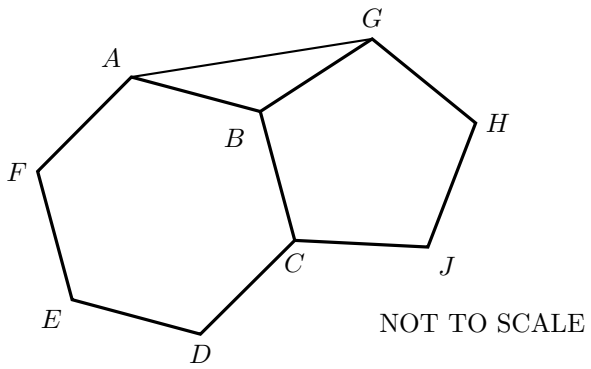
..... ° **[2]**

(b) The reflex angle CBP is 102°.

Calculate the value of n, the number of sides of the regular polygon $ABCDE\ldots$.

$n =$ ..... **[4]**

10 The diagram shows a regular hexagon $ABCDEF$ and a regular pentagon $BCGHJ$ sharing a common side BC.
Vertices A and G are joined by a straight line.



(a) Calculate the size of

(i) interior angle ABC of the regular hexagon,

Angle $ABC =$ ..... ° **[2]**

(ii) interior angle CBG of the regular pentagon.

Angle $CBG =$ ..... ° **[1]**

(b) Calculate the size of angle BAG.

Angle $BAG =$ ..... ° **[3]**

(c) An irregular quadrilateral has interior angles $(2x + 10)°$, $(3x - 15)°$, $(x + 25)°$, and $100°$.
Calculate the size of the largest interior angle of this quadrilateral.

..... ° **[3]**

11 Answer the following questions about polygons and angles.

(a) An irregular polygon has n sides.
Three of the interior angles are each $140°$.
The remaining $(n - 3)$ interior angles are each $165°$.

Work out the value of n.

$n =$ ..... **[4]**

(b) In triangle ABC, the angles are $\angle A = a°$, $\angle B = b°$ and $\angle C = c°$.
It is given that $c - b = a$.

Prove that ABC is a right-angled triangle.

[2]

(c) Polygon P is a regular polygon with 12 sides.
Polygon Q is a regular polygon with k sides.
The exterior angle of Polygon Q is half the size of the exterior angle of Polygon P.
Polygon P and Polygon Q share a common edge AB.

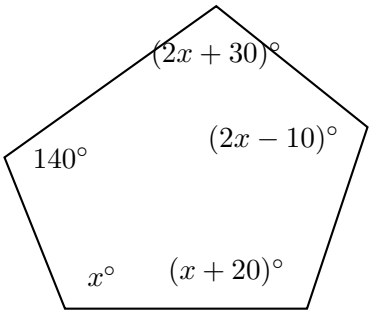
(i) Find the value of k.

$k =$ ..... **[2]**

(ii) Work out the size of the angle at vertex B that lies outside both polygons.

..... ° **[2]**

12 The diagram shows an irregular pentagon.
The interior angles of the pentagon are $x°$, $(x + 20)°$, $(2x - 10)°$, $(2x + 30)°$ and $140°$.



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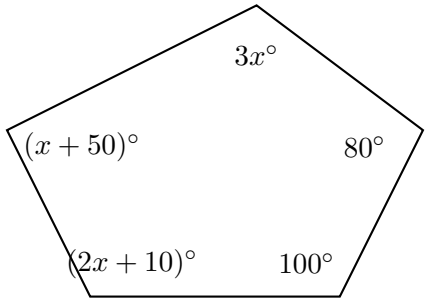
(a) Calculate the value of x.

$x =$ ..... **[3]**

(b) Find the size of the largest interior angle of the pentagon.

..... ° **[2]**

13 The diagram shows an irregular pentagon.



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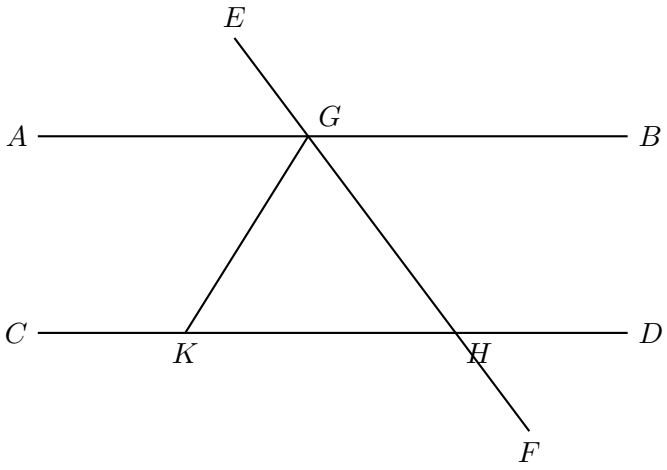
(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Another regular polygon has an interior angle that is 4 times the size of its exterior angle.
Calculate the number of sides of this polygon.

..... **[3]**

14 In the diagram, the line AB is parallel to the line CD.
The transversal EF intersects AB at G and CD at H.
The point K lies on CD.



(a) Angle $EGB = (5x + 10)^\circ$ and angle $GHD = (7x - 30)^\circ$.
Find the value of x.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

(b) Angle $HGK = 35^\circ$.
Calculate angle GKH.

..... $^\circ$ **[3]**

15 **(a)** This question is about the interior and exterior angles of polygons.

(i) The interior angles of a hexagon are $x°$, $2x°$, $(x + 30)°$, $(2x + 10)°$, $(3x - 20)°$, and $160°$.
Work out the value of x.

$x =$ ..... **[3]**

(ii) A regular polygon has an interior angle that is 6.5 times its exterior angle.
Calculate the number of sides of this polygon.

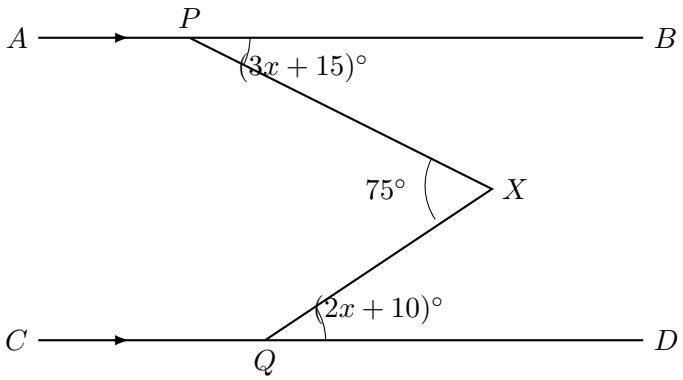
..... **[3]**

(iii) Two regular polygons, P and Q, have m sides and n sides respectively.
The sum of their exterior angles is $54°$.
The difference between their interior angles is $18°$.
Find the value of m and the value of n.

$m =$ ..... , $n =$ ..... **[5]**

16 **(a)** In the diagram, the line AB is parallel to the line CD.
X is a point between AB and CD.
The point P is on AB and the point Q is on CD.
Angle $BPX = (3x + 15)°$, angle $DQX = (2x + 10)°$ and angle $PXQ = 75°$.

Find the value of x.



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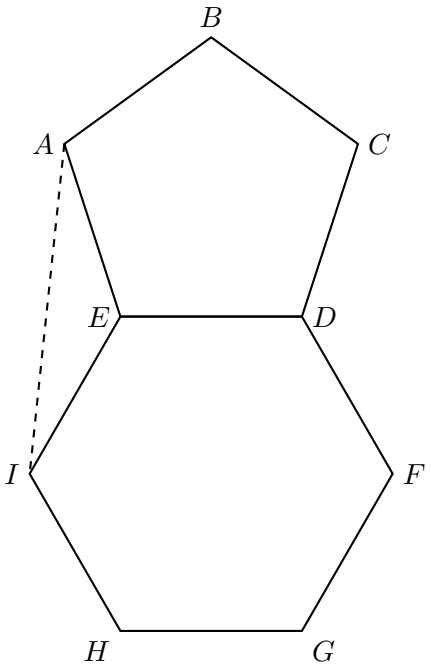
$x =$ ..... **[3]**

(b) Polygon P is a regular polygon with n sides.
Polygon Q is a regular polygon with $3n$ sides.
The size of an interior angle of polygon Q is 16° greater than the size of an interior angle of polygon P.

Calculate the value of n.

$n =$ ..... **[4]**

17 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge ED.



(a) Calculate angle CDE.

Angle $CDE =$ ..... ° **[2]**

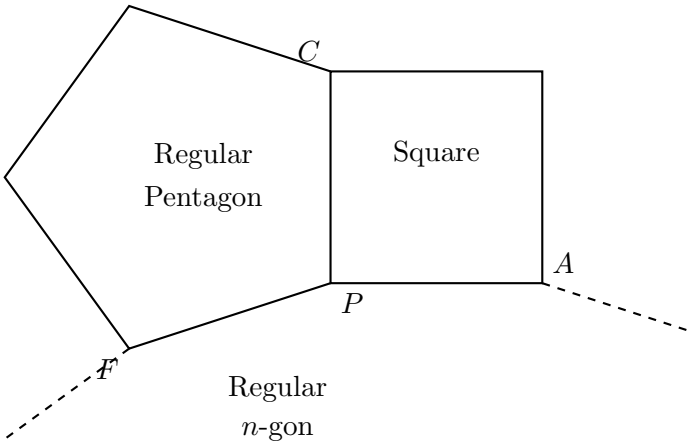
(b) Calculate angle AEI.

Angle $AEI =$ ..... ° **[2]**

(c) Calculate angle EAI.

Angle $EAI =$ ..... ° **[3]**

18 The diagram shows three regular polygons meeting at the point P.
The polygons are a square, a regular pentagon, and a regular polygon with n sides.
The square shares the side PC with the pentagon and the side PA with the n-gon.
The pentagon shares the side PF with the n-gon.
They meet with no gaps and do not overlap.



(a) Calculate the interior angle of a regular pentagon.

..... ° **[2]**

(b) Find the value of n.

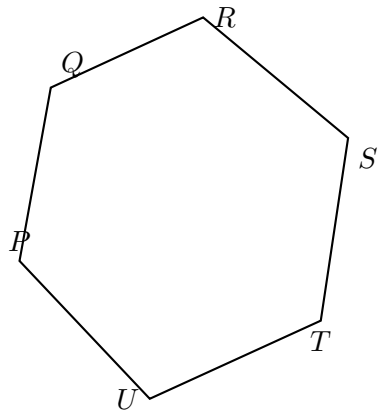
$n =$ ..... **[3]**

(c) The vertices A and F of the n-gon are joined by a straight line.
Calculate angle PFA.

..... ° **[2]**

19 The diagram shows an irregular hexagon $PQRSTU$.
The interior angles of the hexagon are:

- $\angle P = (2x + 15)^\circ$
- $\angle Q = (3x - 20)^\circ$
- $\angle R = (2x + 35)^\circ$
- $\angle S = (4x - 45)^\circ$
- $\angle T = (x + 50)^\circ$
- $\angle U = (3x + 10)^\circ$



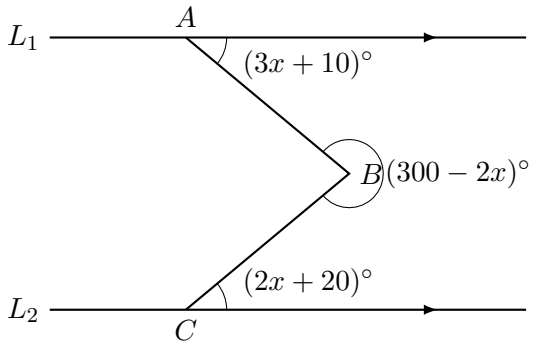
(a) Calculate the value of x.

$x =$ ..... **[4]**

(b) Determine whether PQ is parallel to TS, showing clear geometrical working to support your answer.

..... **[3]**

20 The diagram shows two parallel lines, L_1 and L_2.
The point A lies on L_1 and the point C lies on L_2.
The point B lies between the two parallel lines.
The acute angle between L_1 and AB is $(3x + 10)°$.
The acute angle between L_2 and CB is $(2x + 20)°$.
The reflex angle ABC is $(300 - 2x)°$.



(a) Show that $x = 10$.

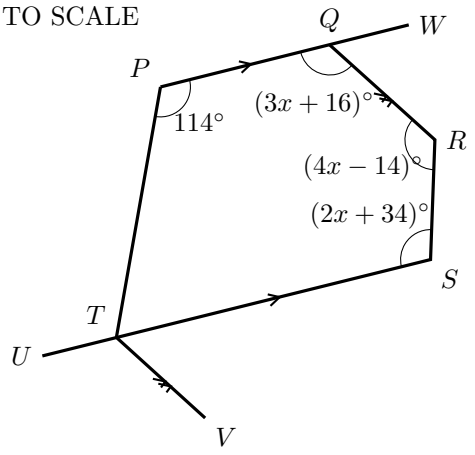
[4]

(b) The points A, B, and C form a triangle ABC.
Given that $AB = BC$, calculate the angle BCA.

Angle $BCA =$ ..... ° **[3]**

21 The diagram shows an irregular pentagon $PQRST$.
The side PQ is extended to W and the side ST is extended to U.
PW is parallel to US, and QR is parallel to TV.
Angle $QPT = 114°$, angle $PQR = (3x + 16)°$, angle $QRS = (4x - 14)°$, and angle $RST = (2x + 34)°$.

NOT TO SCALE



(a) Find the size of angle PTS and give a geometrical reason for your answer.

Angle $PTS =$ ..... , Reason: ..... **[2]**

(b) Show that $x = 36$.

[3]

(c) Calculate the size of

(i) angle PTU,

Angle $PTU =$ ..... ° **[1]**

(ii) angle STV.

Angle $STV =$ ..... ° **[2]**

(d) A regular polygon has n sides.
Each interior angle of this regular polygon is 11 times the size of each exterior angle.
Calculate the value of n.

$n =$ ..... **[3]**